

ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Generative Models

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SELECTED PUBLICATIONS

1. H. Ying, **Allen Tu**, J. Wu, T. Goldstein, and M. Zwicker, ‘UniBRep: Learning Unified Geometry and Topology for Image-conditioned B-Rep Generation’. *Under review*, 2026.
2. **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeedDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *CVPR*, 2026.
3. P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *CVPR*, 2026.
4. **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *FG*, 2026 (**Oral Presentation**).
5. A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *CVPR*, 2025.
6. A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *CVPR*, 2025.

* denotes equal contribution.

EDUCATION

University of Maryland, College Park

Ph.D. Student, Computer Science

College Park, MD

January 2025 – May 2028

- **Advisors:** Professor Tom Goldstein, Professor Matthias Zwicker

- **Collaborators:** Professor Vishal M. Patel, Professor Rama Chellappa

B.S/M.S. in Computer Science, Minor in Statistics

August 2019 – December 2024

RESEARCH EXPERIENCE

Amazon

Applied Science Intern

Bellevue, WA

June 2026 – Present

- Researching methods for constructing metric-scale, photorealistic simulation environments for training embodied AI in Amazon’s Last Mile Delivery and Automation organization
- Creating scalable production pipelines for reconstructing PhysX-ready 3D digital twins from real-world sensor data passively collected by autonomous delivery platforms
- Investigating feed-forward reconstruction models and diffusion-based techniques to bridge Real2Sim and Sim2Real domain gaps across 3D modeling and perception for autonomous systems

University of Maryland Institute of Advanced Computer Studies

Graduate Research Assistant

College Park, MD

August 2023 – Present

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over 10× fewer primitives while preserving visual fidelity [2, 5, 6]
- Introduced uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [3]

Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2025 – January 2026

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [4]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at $1e-3$ FMR using an encoder trained solely on public data [4]

Computer Vision Research Intern

May 2024 – August 2024

- Achieved 0.86 correlation with ground-truth probe-to-gallery similarity and state-of-the-art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [4]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 12%

Computer Vision Research Co-op

January 2023 – August 2023

- Implemented a mixed-voting ensemble that improved open-set search performance by 24% for face recognition, 39% for body recognition, and 49% for fused multimodal biometric recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52% in severe operating conditions

Computer Vision Research Intern

June 2022 – August 2022

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

University of Maryland Department of Computer Science

Undergraduate Researcher

College Park, MD

January 2021 – December 2022

- Analyzed 2.3 million dataset-visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [7]

RESEARCH PROJECTS

IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)

University of Maryland Institute of Advanced Computer Studies

August 2023 – Present

Unconstrained 3D reconstruction and novel view synthesis in challenging environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [2, 5, 6]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [3]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [5]

IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)

Systems & Technology Research

June 2022 – January 2026

Multimodal, opportunistic fusion of face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment [4]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling
- Style-Based Clothing Transfer for Robust Body Representation Learning

HONORS & AWARDS

FG 2026 Doctoral Consortium: *Selective Computation for Reliable and Scalable Vision Systems*

University of Maryland Travel Awards: Jacob K. Goldhaber Travel Grant, CMSC PhD Travel Grant, ICSSA

SERVICE

SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models

Workshop Co-organizer

CVPR 2026

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing \$1K in industry sponsorship, and moderating keynote and Q&A sessions

Reviewer: IEEE TVCG, SIGGRAPH Asia, IEEE TBIOM, CVPR Workshops, FG, IEEE TCSVT

Conference Service: Presented 5 accepted papers (1 oral, 4 posters) on behalf of authors unable to attend FG 2026

TEACHING EXPERIENCE

Peer Research Mentor

College Park, MD

FIRE: Capital One Machine Learning

January 2021 – December 2022

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

TECHNICAL SKILLS

Languages: Python, CUDA, C/C++, C#, Java, SQL, Shell

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn, COLMAP, Open3D, PhysX

Specializations: 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Diffusion Models, Multimodal Biometrics, Uncertainty Quantification, Vision Transformers, GPU Programming